

Lecture 13

DTOs, Validation and API Design

Continuation of the layered Expense Tracker

A DTO is a boundary, not a duplicate entity for decoration.

- Separate public API contracts from internal entities.
- Validate request shape and business rules.
- Return predictable errors and status codes.

Domain: internal entity.

Application: reusable rules.

Infrastructure: persistence.

Presentation: DTOs, mapping and HTTP validation.

Project: host.

Request DTO controls allowed input.

Response DTO controls exposed output.

ValidationProblem is consistent for clients.

Routes use nouns and HTTP verbs.

- Read the runnable CodeExamples project first.
- Run the Project and identify where each responsibility lives.
- Change one implementation detail without breaking the Application layer.
- Write a short explanation of the boundary you protected.

- Apply the same decision to the student's chosen domain.
- Keep a screenshot or short note for the evolution portfolio.
- Use the database-session model and constraints.
- Prepare one demonstration scenario for the final defense.